25900801 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR /IMPROVEMENT/ REAPPEARANCE EXAMINATION NOVEMBER 2025

FIRST SEMESTER

Discipline Specific Core Course (DSC) - MG1DSCECC100, MG1DSCECT100 - EMERGING ELECTRONICS

(2024 ADMISION ONWARDS)

Duration: 1.5 Hours **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer **All** Questions

Each Question carries **1** mark

1. What does the term “load” refer to in a voltage regulator circuit? [K] / [CO1]

   a). The Zener diode itself | b). The power supply | c). The component drawing current | d). The input voltage source

2. What is the relationship between the number of turns in the primary and secondary coils of a transformer and the voltage? [U] / [CO1]

   a). Directly proportional | b). Inversely proportional | c). Unrelated | d). Exponentially related

3. What is the unit of capacitance? [K] / [CO1]

   a). Farad | b). Henry | c). Ohm | d). Volt

4. In a common-emitter amplifier configuration, which terminal is the input? [K] / [CO2]

   a). Emitter | b). Base | c). Collector | d). Ground

5. What is the output of a negative clipper when the input is below the clipping level? [U] / [CO2]

a). Zero | b). Input voltage | c). Clipped voltage | d). Negative voltage

6. ------- cannot be fabricated on IC [K] / [CO2]

a). Resistor | b). Transistor | c). Transformer | d). Capacitor

7. Which component is used in automatic street lighting? [K] / [CO3]

a). LDR | b). Thermistor | c). Microphone | d). Ultrasonic sensor

8. In which devices are thermistors commonly used? [K] / [CO3]

a). Sound systems | b). Temperature control circuits | c). Light detectors | d). Pressure sensors

9. Two resistors, R1 and R2 , are connected in series. What is the equivalent resistance (Rt)? [K] / [CO1]

a). R1 + R2 | b). R1 × R2 | c). R1 × R2 / (R1 + R2) | d). (R1 + R2) /R1*R2

10. How does a mechanical relay control a high-power circuit? [K] / [CO3]

a). By direct contact | b). By magnetic induction | c). By electronic control | d). By generating heat

[10x1 = 10]

Part B

Short answer Type Questions

Answer any **4** Questions

Each Question carries **5** marks

11. Explain forward current and reverse current in a diode. [U] / [CO1]

12. Extract the benefits of biometric authentication. [K] / [CO1]

13. What are the main characteristics of P-type semiconductors [U] / [CO1]

14. In a clamper circuit, what role does the capacitor play? [U] / [CO2]

15. What do you understand by avalanche breakdown? [K] / [CO1]

16. How does a fuse provide overcurrent protection? [U] / [CO3]

[4x5=20]

Part C

Essay Type Questions

Answer any **2** Questions

Each Question carries **10** marks

17.	Explain the basic principle of resistors and capacitors. Briefly explain its classification	[U] / [CO1]
18.	Explain the V-I Characteristics of JFET	[U] / [CO2]
19.	Explain the working principle of a half-wave rectifier and discuss its limitations and applications.	[U] / [CO2]
20.	What are the advantages of using a DPST switch over an SPST switch?	[QNSMARTI:T:B20] / [CO3]

[2x10 =20]